UNSW Course Outline

ZZSC9001 Foundations of Data Science - 2026

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General Course Information

Course Code : ZZSC9001
Year : 2026
Term : Hexamester 2
Teaching Period : KF
Is a multi-term course? : No
Faculty : Faculty of Science
Academic Unit : School of Mathematics & Statistics
Delivery Mode : Online
Delivery Format : Standard
Delivery Location : Kensington
Campus : Sydney
Study Level : Postgraduate
Units of Credit : 6

Useful Links

Handbook Class Timetable

Course Details & Outcomes

Course Description

This is an intensive course that covers the fundamentals of Data Science as it is applied in Computer Science, Economics, Mathematics and Statistics. The course will introduce topics such as databases, data analytics, data mining, Bayesian statistics, statistical software,

econometrics, machine learning, and business forecasting. The course also introduces the Data Science and Decisions program, indicating the relevance of the courses that follow in the program (including electives) and their place in data science and its applications.

This course includes lectures, tutorials, and workshops and is delivered completely online. The course will take the flipped classroom approach. Students are assigned the “homework” of watching video lectures and reading materials relevant to the online lecture session. During each session, a recap and overview of the content will be given and students will practice what they’ve learned through examples and tutorial problems.

Course Aims

The aim of this course is to provide a broad overview of modern Data Science techniques that are broadly applicable in Computer Science, Economics, Mathematics and Statistics. This course provides a platform for further studies in Data Science. The course activities and content will also provide a foundation to analyse and address problems that data scientists face in the modern world.

Course Learning Outcomes

Course Learning Outcomes
CLO1 : Apply probability rules in a given setting to calculate key quantities.
CLO2 : Use key theoretical tools to explore the properties of random variables.
CLO3 : Apply key methods of statistical inference in applied settings.
CLO4 : Use R/RStudio to perform statistical computations and simulations.
CLO5 : Apply various data visualisation tools, perform regression analysis and draw casual inference from data.
CLO6 : Apply fundamental data science techniques and tools, including machine learning, Naïve Bayes classification, Decision trees, K-Nearest Neighbour, unsupervised learning and neural networks.

Course Learning Outcomes	Assessment Item
CLO1 : Apply probability rules in a given setting to calculate key quantities.	• Statistics Quiz • Final Exam
CLO2 : Use key theoretical tools to explore the properties of random variables.	• Statistics Quiz • Final Exam
CLO3 : Apply key methods of statistical inference in applied settings.	• Statistics Quiz • Final Exam
CLO4 : Use R/RStudio to perform statistical computations and simulations.	• Economics Report • Statistics Quiz • Final Exam
CLO5 : Apply various data visualisation tools, perform regression analysis and draw casual inference from data.	• Economics Report • Final Exam
CLO6 : Apply fundamental data science techniques and tools, including machine learning, Naïve Bayes classification, Decision trees, K-Nearest Neighbour, unsupervised learning and neural networks.	• Computer Science Assignment • Final Exam

Learning and Teaching Technologies

Moodle - Learning Management System | Blackboard Collaborate | EdStem

Learning and Teaching in this course

The course contains a variety of resources and activities that are carefully designed to enhance your learning.

Some activities require you to work and think alone, by reading some text, listening to a recording or watching a video. You might be asked to engage with the material and explore interactive

elements by clicking to reveal content, to help you better absorb and process the concepts. Some activities require you to produce work of your own. You might be answering a question, writing code to solve a problem, or posting to a forum, for example. Some activities are assessment tasks, which have been carefully designed to measure how well you have achieved the learning outcomes of the course. Typically, you will get feedback on your work, either from yourself (by checking your work with models that are provided), or from an automatic marking process, or from your peers, or from your teacher.

You also have access to a variety of ways to communicate with your peers and with the teaching staff. The general discussion forums are a place for you to ask and answer questions, to interact with your peers, and to be challenged by your teachers. Getting involved in these forums will enhance your learning experience and make it more enjoyable. Your course may include Webinars, which provide an opportunity to hear directly from your Online Lecturers and ask questions in real time. All webinars are recorded so you can access them at any time. Online Lecturers are available for consultations and will post information about how to access consultations on the course website. You can also contact your Online Lecturer by email using the email address in the teaching staff section of this outline.

It is up to you how much work you do. The more time and effort that you can dedicate to the course, the better will be your learning and your results.

Assessments

Assessment Structure

Assessment Item	Weight	Relevant Dates
Statistics Quiz Assessment Format: Individual	15%	Due Date: Monday of Weeks 2 and 3 by 10am Sydney time
Economics Report Assessment Format: Individual	15%	Due Date: Monday of Week 5 by 10am Sydney time
Computer Science Assignment Assessment Format: Individual	15%	Due Date: Friday of Week 6 by 5pm Sydney time
Final Exam Assessment Format: Individual	55%	Due Date: Wednesday of Week 7 at 12pm (noon) Sydney time

Assessment Details

Statistics Quiz

Assessment Overview

There are two parts to this quiz. This quiz assesses your understanding of the material covered from Weeks 1 to 2.

The first part of this quiz is due at the beginning of Week 2, and the second part is due at the beginning of Week 3. Specific details of the assessment will be available during the term.

You will receive feedback in the form of a mark with detailed worked solutions for each question after the due date.

The rationale of this assessment is to give you feedback on your progress and mastery of the material, and to obtain measures of your progress towards the stated learning outcomes.

Course Learning Outcomes

- CLO1 : Apply probability rules in a given setting to calculate key quantities.
- CLO2 : Use key theoretical tools to explore the properties of random variables.
- CLO3 : Apply key methods of statistical inference in applied settings.
- CLO4 : Use R/RStudio to perform statistical computations and simulations.

Assessment Length

2 hours

Assignment submission Turnitin type

This is not a Turnitin assignment

Assessment AI Mode

Simple Editing Assistance

In completing this assessment, you are permitted to use standard editing and referencing functions in the software you use to complete your assessment. These functions are described below. You must not use any functions that generate or paraphrase passages of text or other media, whether based on your own work or not.
If your Convenor has concerns that your submission contains passages of AI-generated text or media, you may be asked to account for your work. If you are unable to satisfactorily demonstrate your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.

For more information on Generative AI and permitted use please see here.

Economics Report

Assessment Overview

This is a written assessment that will assess your ability to apply some of the concepts covered in this course to a real-world business problem by performing regression analysis on a dataset and applying various data visualization tools. You are expected to write a short report of under 1,000 words

The assessment is due approximately around the start of Week 5. Specific details of the assessment will be available during the term.
You will receive feedback in the form of a marked rubric and individualised comments.

Course Learning Outcomes

- CLO4 : Use R/RStudio to perform statistical computations and simulations.
- CLO5 : Apply various data visualisation tools, perform regression analysis and draw casual inference from data.

Assessment Length

1000 words

Assignment submission Turnitin type

This assignment is submitted through Turnitin and students can see Turnitin similarity reports.

Assessment AI Mode

Assistance with Attribution

This assessment requires you to write/create a first iteration of your submission yourself. You are then permitted to use generative AI tools, software or services to improve your submission in the ways set out below.
Any output of generative AI tools, software or services that is used within your assessment must be attributed with full referencing.
If outputs of generative AI tools, software or services form part of your submission and are not appropriately attributed, your Convenor will determine whether the omission is significant. If so, you may be asked to explain your submission. If you are unable to satisfactorily demonstrate your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.
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Computer Science Assignment

Assessment Overview

This assessment requires you to answer short questions covering topics from Weeks 5 to 6.

The assessment is due approximately around the end of Week 6. Specific details of the assessment will be available during the term.

The rationale of this assessment is to give you feedback on your progress and mastery of the material and to obtain measures of your progress toward the stated learning outcomes. You will receive feedback in the form of a marked rubric and individualised comments.

Course Learning Outcomes

- CLO6 : Apply fundamental data science techniques and tools, including machine learning, Naïve Bayes classification, Decision trees, K-Nearest Neighbour, unsupervised learning and neural networks.

Assessment Length

1 hour

Assignment submission Turnitin type

This is not a Turnitin assignment

Assessment AI Mode

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Final Exam

Assessment Overview

The final exam is designed to summarise your learning and problem-solving skills on all topics

delivered across all weeks of the term, including material from lectures, tutorials and workshops. The exam is typically 3hrs long and consists of MCQ, short numerical and short answer responses - details will be confirmed during the course. The examination will be delivered online and will occur during the official university examination period.

The rationale of this assessment is to give you feedback on your progress and mastery of the material, and to obtain measures of your progress towards the stated learning outcomes.

Additional feedback is available through consultation with the course convenor.

Course Learning Outcomes

- CLO1 : Apply probability rules in a given setting to calculate key quantities.
- CLO2 : Use key theoretical tools to explore the properties of random variables.
- CLO3 : Apply key methods of statistical inference in applied settings.
- CLO4 : Use R/RStudio to perform statistical computations and simulations.
- CLO5 : Apply various data visualisation tools, perform regression analysis and draw casual inference from data.
- CLO6 : Apply fundamental data science techniques and tools, including machine learning, Naïve Bayes classification, Decision trees, K-Nearest Neighbour, unsupervised learning and neural networks.

Assessment Length

2 hours

Assignment submission Turnitin type

This is not a Turnitin assignment

Assessment AI Mode

Simple Editing Assistance

In completing this assessment, you are permitted to use standard editing and referencing functions in the software you use to complete your assessment. These functions are described below. You must not use any functions that generate or paraphrase passages of text or other media, whether based on your own work or not.
If your Convenor has concerns that your submission contains passages of AI-generated text or media, you may be asked to account for your work. If you are unable to satisfactorily demonstrate your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.
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General Assessment Information

UNSW standard late submission penalty of:

- *5% per day, for all assessments where a penalty applies, capped at two days (48 hours) from the assessment deadline, after which a student cannot submit an assessment, and no permitted variation*

Grading Basis

Standard

Course Schedule

Teaching Week/Module	Activity Type	Content
Week 0 : 9 March - 15 March	Online Activity	Orientation week
Week 1 : 16 March - 22 March	Online Activity	Probability and randomness
Week 2 : 23 March - 29 March	Online Activity	Inferential statistics
Week 3 : 30 March - 5 April	Online Activity	Collecting and displaying data
Week 4 : 6 April - 12 April	Online Activity	Data analysis and econometrics
Week 5 : 13 April - 19 April	Online Activity	Binary trees, B-trees and decision trees
Week 6 : 20 April - 26 April	Online Activity	Big data and its challenges

Attendance Requirements

Students are strongly encouraged to attend all classes and review lecture recordings.

Course Resources

Course Evaluation and Development

Toward the end of the hexamester you will be asked to give feedback about the course, via UNSW's MyExperience survey. Your feedback will be used, along with feedback from other stakeholders, to help improve the course. You can also contact your Course Convenor any time you have suggestions or other feedback.

Staff Details

Position	Name	Email	Location	Phone	Availability	Equitable Learning Services Contact	Primary Contact
Program director	Donna Mary Salopek					No	No
Convenor	Ruyi Liu					Yes	Yes
	Martin Eftimoski					No	No